Chapter 7

A Primer on Asymptotic Normality

The way in which results involving asymptotic normality are presented in the literature can differ from reference to reference, and the most appropriate representation typically depends on the specific form of technical conditions under which those results are developed. In what follows we have two objectives, to present results in notation that can be related to notation used the references cited, and to make clear the general and common structure involved in results connected with asymptotic normality. Let $\boldsymbol{\theta}$ denote a true parameter value with q individual components, and let $\hat{\boldsymbol{\theta}}_n$ denote an estimator of $\boldsymbol{\theta}$ based on n random variables. Typically, $\hat{\boldsymbol{\theta}}_n$ will have been obtained by maximizing some objective function $\ell^{(n)}(\boldsymbol{\theta}|\boldsymbol{y})$. The objective function $\ell^{(n)}(\boldsymbol{\theta}|\boldsymbol{y})$ may be a log likelihood, a log quasi-likelihood, a log composite likelihood or some other objective function connected with an unbiased estimating function. These are topics that are covered in later chapters in these notes. For now, simply accept that there are some functions that we

can construct from observed data and unknown parameters that, when maximized in the parameter values, provide estimates that may have discernible properties. The most notable among such properties is asymptotic normality.

7.1 Asymptotic Context

The production of asymptotic results always involves what is known as the *asymptotic context* within which results are formulated. For models that involve independence and no grouping of variables this is not much of an issue. Sample size increases by obtaining more and more independent realizations of whatever model is under consideration. Many of the estimation methods for which we will be interested in asymptotic results, however, are motivated at least in part through the presence of additional structure in the collection of random variables involved in a model. For example, consider a longitudinal data structure in which we have m_i random variables for each of $i = 1, \ldots, n$ individuals. Here, the total number of random variables involved in the model can grow large by having $m_i \to \infty$ for some or all i, but with n fixed, by having $n \to \infty$ but with a fixed and common value $m_i = m$, or by having both $m_i = m \to \infty$ and $n \to \infty$ in some specified manner, such as with $n/m = \lambda < \infty$ for a fixed λ, or with $n/m \to \infty$ as well as each of m and n individually. Similarly, for a spatial problem defined on a finite index random field such as a lattice, there are several ways that overall sample size can grow without bound. One way that sample size can grow large is if a lattice defined by a fixed spacing and configuration of locations simply expands without bound. This is often called the *expanding lattice* asymptotic context. Another way for sample size to increase is if the number of independent realizations of a process on a given fixed finite

lattice increases. This is called the *repeating lattice* context. In principle, this might occur if a given spatial region is observed over and over again in time, with temporal spacing great enough to result in independent realizations and under the assumption that the processes involved in producing response values has not changed over time.

To a degree, physical reality does not need to match an asymptotic context in order for asymptotic results to be applied. That is, asymptotic results are about the behavior of the model and estimators as sample size increases, not about real situations. For example, just because there is an ocean or lake on one side of a spatial study region used to observe a terrestrial response does not mean that the expanding lattice context is meaningless. At the same time, asymptotic results are directly applicable only if the overall number of observations is large enough for limiting results to provide reasonable approximations to the necessarily finite problem under consideration. Thus, in the immediately preceding example, the observed spatial lattice must be large enough for the hypothetical infinitely expanding lattice results to be meaningful. Similarly, if the asymptotic context for a problem involving groups of random variables is that the number of groups grows large for a fixed number of variables within each group, then there must be enough groups observed, even if it is physically impossible that this number could grow large without bound.

Theoretical results are typically easier to obtain for an asymptotic context under which one obtains an increasing number of independent replicates of multivariate observations than they for an asymptotic context in which the dimension of a multivariate distribution increases without bound. This corresponds to settings in which the number of groups of random variables increases, be those multiple measurements on individuals or other objects,

or multiple observations of a fixed spatial region or a process of a fixed temporal extent. The context in which one obtains larger and larger collections of random variables for a single realization of a stochastic process, such as an expanding spatial lattice or a time series of increasing length, typically requires that additional structure be imposed on a model in order to obtain repeated realizations of the same statistical or probabilistic behavior.

7.2 Forms of Asymptotic Normality

There are a number of different forms, and hence notations, that statements of asymptotic normality can take. These forms differ primarily in terms of what is used as sequences of normalizing matrices for asymptotic normality, and whether they are or are not stochastic, and whether they do or do not converge to something and, if so, what that something is.

A general form for the expression of asymptotic normality of an estimator $\hat{\boldsymbol{\theta}}_n$ is

$$\mathcal{I}_n^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) \xrightarrow{\mathcal{L}} \text{MVN}(\mathbf{0}_q, I_q) \text{ as } n \to \infty, \tag{7.1}$$

where $\mathcal{L}$ denotes convergence in law or distribution, MVN denotes a multivariate normal distribution, $\mathbf{0}_q$ is a $q-$vector of $0's$, and I_q is the $q \times q$ identity matrix with $1's$ on the diagonal and $0s$ elsewhere. Whatever the form of $\mathcal{I}_n$ involved, the implication of this result is that, for suitably large n, we may behave as if $\hat{\boldsymbol{\theta}}_n$ has a multivariate normal distribution with mean $\boldsymbol{\theta}$ and covariance matrix $\mathcal{I}_n^{-1}$. In particular, $(1-\alpha)\,100\%$ confidence intervals for individual elements of $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_q)$ may be formed as

$$\hat{\theta}_k \pm z_{1-\alpha/2}\,[\hat{i}^{k,k}]; \quad k = 1, \ldots, q, \tag{7.2}$$

where $\hat{\theta}_k$ is the kth element of the estimator $\hat{\boldsymbol{\theta}}_n$, $\hat{i}^{k,k}$ is the kth diagonal

element of the estimated version of $\mathcal{I}_n^{-1}$, and $z_{1-\alpha/2}$ is the $(1-\alpha/2)$ quantile of a standard normal distribution.

A source of potential confusion for those wishing to make use of asymptotic results in applications is that there is not necessarily a unique sequence of matrices $\mathcal{I}_n$ that will cause (7.1) to hold. Different expressions giving asymptotic normality arise from differences in the forms chosen for the sequence of normalizing matrices $\mathcal{I}_n$ and the behavior of these forms as n grows large. In the general representation of asymptotic normality, there is no particular behavior required of the sequences $\{\hat{\boldsymbol{\theta}}_n : n = 1, \ldots, \}$ and $\{\mathcal{I}_n : n = 1, \ldots, \}$ other than that (7.1) hold. Technically, even $\hat{\boldsymbol{\theta}}_n$ need not converge in probability to $\boldsymbol{\theta}$, although it would seem difficult to assert that anything useful is being learned about the possible value of $\boldsymbol{\theta}$ if $\hat{\boldsymbol{\theta}}_n$ is not a consistent sequence of estimators, and we have adopted the convention here that any estimator to be considered is consistent. Given this, each element of the sequence of normalizing matrices $\mathcal{I}_n$ must grow large without bound as $n \to \infty$, will typically depend on the value of the parameter $\boldsymbol{\theta}$, and may be stochastic in that they are also functions of $\boldsymbol{Y}$.

Suitable choices for the sequence of matrices $\mathcal{I}_n$ in (7.1) largely depends on three considerations: (1) what can be shown to exist, (2) what can be explicitly identified, and (3) what can be computed on the basis of observed data. This last consideration is, of course, crucial for making practical use of a result in application. Some of the types of $\mathcal{I}_n$ and the typical notation used for each will now be given, and an example based on maximum likelihood estimation will provide more detail on possible specific forms of these types in the next section.

1. It may be that we can choose $\mathcal{I}_n = n\,\Sigma^{-1}(\boldsymbol{\theta})$, where $\Sigma^{-1}(\boldsymbol{\theta})$ is a non-

stochastic positive definite matrix that does not depend on n. Then (7.1) is often written as

$$n^{1/2}[\Sigma^{-1}(\boldsymbol{\theta})]^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) \xrightarrow{\mathcal{L}} MVN(\mathbf{0}_q,\, I_q). \tag{7.3}$$

Because $\Sigma^{-1}(\boldsymbol{\theta})$ is non-stochastic and does not depend on n one could also write

$$n^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) \xrightarrow{\mathcal{L}} MVN(\mathbf{0}_q,\, \Sigma(\boldsymbol{\theta})). \tag{7.4}$$

2. Another form for the normalizing matrices that may be appropriate is $\mathcal{I}_n = c_n\, \mathcal{H}_n(\boldsymbol{\theta})$, where $\mathcal{H}_n(\boldsymbol{\theta})$ is a non-stochastic matrix depending on n but that converges (elementwise and in the ordinary sense) to a positive definite matrix $\Sigma^{-1}(\boldsymbol{\theta})$. Here, c_n; $n \geq 1$ denotes a sequence of constants that depend on n. Usually, $c_n = n$ but this is neither guaranteed nor necessary. Relative to the three considerations listed previously, this form for the $\mathcal{I}_n$ may be used when we can demonstrate the existence of $\Sigma^{-1}(\boldsymbol{\theta})$ but explicitly identify only the $\mathcal{H}_n(\boldsymbol{\theta})$. Here, (7.1) might be written as,

$$c_n^{1/2}[\mathcal{H}_n(\boldsymbol{\theta})]^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) \xrightarrow{\mathcal{L}} MVN(\mathbf{0}_q,\, I_q). \tag{7.5}$$

It would no longer be proper to put $\mathcal{H}_n^{-1}(\boldsymbol{\theta})$ on the right-hand side as was done for $\Sigma(\boldsymbol{\theta})$ in (7.4), because of its dependence on n.

3. Another possibility is that $\mathcal{I}_n = c_n \mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ where c_n is a sequence of constants that depend on n and $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ is a stochastic matrix that converges in probability to a non-stochastic positive definite matrix $\Sigma^{-1}(\boldsymbol{\theta})$. As for the previous situation in item (2), often we will have $c_n = n$, but this is not necessary and will not always be true. In these situations we might write (7.1) as

$$c_n^{1/2}[\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})]^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) \xrightarrow{\mathcal{L}} MVN(\mathbf{0}_q,\, I_q). \tag{7.6}$$

Similar to the use of $\mathcal{H}_n(\boldsymbol{\theta})$ in item (2), this type of choice for the $\mathcal{I}_n$ arises when we are able to demonstrate the existence of $\Sigma^{-1}(\boldsymbol{\theta})$ but identify only the $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$. Although not exhaustive of the situations under which this choice for the form of the $\mathcal{I}_n$ is possible, it is not uncommon that we are in a situation in which we could use result (7.5) with $\mathcal{H}_n(\boldsymbol{\theta}) = E\{\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})\}$, but the expectation is intractable so we use $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ instead.

4. The consistent estimator $\hat{\boldsymbol{\theta}}_n$ may often be substituted for $\boldsymbol{\theta}$ in any of $\Sigma^{-1}(\boldsymbol{\theta})$, $\mathcal{H}_n(\boldsymbol{\theta})$ or $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ to produce a sequence of matrices $\mathcal{I}_n$ in (7.1), without changing the results (7.3), (7.5) or (7.6), as long as these matrices are smooth functions of $\boldsymbol{\theta}$, which holds for many models. The technical meaning of smooth varies a bit depending upon which type of normalizing matrix in which we wish to replace $\boldsymbol{\theta}$ with $\hat{\boldsymbol{\theta}}_n$. With $\hat{\boldsymbol{\theta}}_n$ consistent for $\boldsymbol{\theta}$, the elements of $\Sigma^{-1}(\hat{\boldsymbol{\theta}}_n)$ are consistent estimators of the elements of $\Sigma^{-1}(\boldsymbol{\theta})$ when those elements are continuous functions of $\boldsymbol{\theta}$. With $\hat{\boldsymbol{\theta}}_n$ consistent for $\boldsymbol{\theta}$ and under other conditions sufficient for $\mathcal{H}_n(\boldsymbol{\theta})$ to converge to $\Sigma^{-1}(\boldsymbol{\theta})$, $\mathcal{H}_n(\hat{\boldsymbol{\theta}}_n)$ is a consistent estimator of $\Sigma^{-1}(\boldsymbol{\theta})$ if $\mathcal{H}_n(\boldsymbol{\theta})$ is equicontinuous at $\boldsymbol{\theta}$. This means that we may choose a small distance δ and a neighborhood of $\boldsymbol{\theta}$ (that may depend on δ), call it $B(\boldsymbol{\theta}, \delta)$, such that $||\mathcal{H}_n(\boldsymbol{\theta}') - \mathcal{H}_n(\boldsymbol{\theta})|| < \delta$ for all $\boldsymbol{\theta}' \in B(\boldsymbol{\theta}, \delta)$ when n is large. Similarly, with $\hat{\boldsymbol{\theta}}_n$ consistent for $\boldsymbol{\theta}$ and under other conditions sufficient for $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ to be consistent for $\Sigma^{-1}(\boldsymbol{\theta})$, $\mathcal{J}_n(\hat{\boldsymbol{\theta}}_n, \boldsymbol{Y})$ is a consistent estimator of $\Sigma^{-1}(\boldsymbol{\theta})$ if $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ is stochastically equicontinuous at $\boldsymbol{\theta}$. This means that we may choose a small distance δ and a neighborhood of $\boldsymbol{\theta}$ (that may depend on δ), call it $B(\boldsymbol{\theta}, \delta)$, such that $Pr[||\mathcal{J}_n(\boldsymbol{\theta}', \boldsymbol{Y}) - \mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})|| < \delta$ for all $\boldsymbol{\theta}' \in B(\boldsymbol{\theta}, \delta)] > 1 - \epsilon$ for any

$\epsilon > 0$ when n is large. Under these conditions we would have, in addition to (7.3), (7.5) or (7.6),

$$\begin{aligned} n^{1/2}[\Sigma^{-1}(\hat{\boldsymbol{\theta}}_n)]^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) &\xrightarrow{\mathcal{L}} MVN(\mathbf{0}_q,\, I_q) \\ (c_n)^{1/2}[\mathcal{H}_n(\hat{\boldsymbol{\theta}}_n)]^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) &\xrightarrow{\mathcal{L}} MVN(\mathbf{0}_q,\, I_q) \\ (c_n)^{1/2}[\mathcal{J}_n(\hat{\boldsymbol{\theta}}_n, \boldsymbol{Y})]^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) &\xrightarrow{\mathcal{L}} MVN(\mathbf{0}_q,\, I_q) \end{aligned} \tag{7.7}$$

It may be that results such as (7.3), (7.5) or (7.6) are given and it is then indicated that we would estimate the large-sample covariance matrix of $\hat{\boldsymbol{\theta}}_n$ by substituting $\hat{\boldsymbol{\theta}}_n$ for $\boldsymbol{\theta}$ in $(1/n)\Sigma(\boldsymbol{\theta})$, $(1/c_n)[\mathcal{H}_n(\boldsymbol{\theta})]^{-1}$ or $(1/c_n)[\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})]^{-1}$, but this is only justified if one of the distributional results in (7.7) holds. That is, we are always implicitly using some scaling matrix $\Sigma(\boldsymbol{\theta})$ in (7.4), even if we cannot identify it in explicit form. If we cannot identify it we replace that scaling matrix with sequences of matrices that converge to it, either directly as for $\mathcal{H}_n(\boldsymbol{\theta})$ or in probability as for $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$. We then further replace the parameter $\boldsymbol{\theta}$ in these sequences with a consistent estimator $\hat{\boldsymbol{\theta}}_n$ and, under suitable conditions such as those presented previously, limiting results continue to hold. The distinction between using $\hat{\boldsymbol{\theta}}_n$ as a "plug-in" value for $\boldsymbol{\theta}$ to estimate $\Sigma(\boldsymbol{\theta})$ and the fact that the limiting results (7.5) and (7.6) continue to hold with $\mathcal{H}_n(\hat{\boldsymbol{\theta}}_n)$ or $\mathcal{J}_n(\hat{\boldsymbol{\theta}}_n, \boldsymbol{Y})$ as in (7.7) is important in understanding the final possible form of the $\mathcal{I}_n$ in (7.1), which we now present.

5. We come now to the most confusing and potentially counter-intuitive possibility for choice of the normalizing matrices $\mathcal{I}_n$ in (7.1). It may be possible to identify (almost always stochastic) matrices $\mathcal{I}_n = \mathcal{M}_n(\hat{\boldsymbol{\theta}}_n, \boldsymbol{Y})$ such that (7.1) holds, but not a sequence of constants c_n such that we could take $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y}) = (1/c_n)\mathcal{M}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ and verify that these matri-

ces converge in probability to some positive definite matrix $\Sigma^{-1}(\boldsymbol{\theta})$. The issue here is not whether one could write $\mathcal{I}_n = c_n \mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y}) = c_n(1/c_n)\mathcal{M}_n(\boldsymbol{\theta}, \boldsymbol{Y})$, which is trivially true for any c_n. The issue is whether this can be done and result in convergence in probability of $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y}) = (1/c_n)\mathcal{M}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ to some non-stochastic positive definite matrix $\Sigma^{-1}(\boldsymbol{\theta})$ that does not depend on n. If this is not the case then, although as just indicated it would be possible to develop the result (7.6), that result would be of little value because the device of item (4) in the preceding material may not hold true. That is, with $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y}) = (1/c_n)\mathcal{M}_n(\boldsymbol{\theta}, \boldsymbol{Y})$, (7.6) might hold but, if $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ in (7.6) is not consistent for some $\Sigma^{-1}(\boldsymbol{\theta})$, then the third line of (7.7) would not hold. The dramatic distinction between this case and those presented previously is that a choice of $\mathcal{I}_n$ can be identified such that (7.1) holds, but we may not have even verified the existence of a $\Sigma^{-1}(\boldsymbol{\theta})$ such that (7.3) holds. A representation of asymptotic normality in this situation assumes the general form of (7.1) with $\mathcal{I}_n = \mathcal{M}_n(\hat{\boldsymbol{\theta}}_n, \boldsymbol{Y})$ as,

$$[\mathcal{M}_n(\hat{\boldsymbol{\theta}}_n, \boldsymbol{Y})]^{1/2}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) \xrightarrow{\mathcal{L}} MVN(\mathbf{0}_q,\, I_q). \tag{7.8}$$

The implication in practice is that the covariance matrix of the limiting distribution ($\Sigma(\boldsymbol{\theta})$ in (7.3) or (7.4)) cannot really be estimated because that matrix has not necessarily even been shown to exist. Nevertheless, in principle, confidence intervals for the components of $\boldsymbol{\theta}$ can still be constructed through the use of (7.2) as long as n is sufficiently large and the inverse of $\mathcal{M}_n(\hat{\boldsymbol{\theta}}_n, \boldsymbol{Y})$ can be computed.

7.3 Examples of $\mathcal{I}_n$ in a Likelihood Setting

To illustrate some explicit forms of potential $\mathcal{I}_n$ matrices, consider a problem involving n independent observations of $m-$dimensional random variables with distributions that contain q scalar parameters. We will assume the repeating lattice asymptotic context of Chapter 7.1. For what are called *regular* problems with $\ell^{(n)}(\boldsymbol{\theta}|\boldsymbol{y})$ a true log likelihood, the $\boldsymbol{Y}_i$ identically distributed, and $\hat{\boldsymbol{\theta}}_n$ a maximum likelihood estimator, $\mathcal{I}_n$ could be taken as any of the forms $n\Sigma^{-1}(\boldsymbol{\theta})$ in item 1 of the previous section, $c_n\mathcal{H}_n(\boldsymbol{\theta})$ in item 2, or $c_n\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ in item 3, both of these latter two with $c_n = n$. If the $\boldsymbol{Y}_i$ are not identically distributed, then $\Sigma^{-1}(\boldsymbol{\theta})$ would most likely not be available in closed form, but either of the other two forms might suffice.

If the distributions of the $\boldsymbol{Y}_i$ are identical then $\ell^{(n)}(\boldsymbol{\theta}|\boldsymbol{Y}) = \sum_{i=1}^n \ell_i(\boldsymbol{\theta}|\boldsymbol{Y}_i) = \sum_{i=1}^n \ell(\boldsymbol{\theta}|\boldsymbol{Y}_i)$. If we choose $\mathcal{I}_n = n\Sigma^{-1}(\boldsymbol{\theta})$, then $\Sigma^{-1}(\boldsymbol{\theta})$ is the Fisher (or expected) information matrix for a single observation, a $q \times q$ matrix with jk^{th} element

$$d_{j,k} = E\left(\frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{Y}_i)}{\partial \theta_j}\frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{Y}_i)}{\partial \theta_k}\right) = E\left(-\frac{\partial^2 \ell(\boldsymbol{\theta}|\boldsymbol{Y}_i)}{\partial \theta_j\,\partial \theta_k}\right), \tag{7.9}$$

which are constants that are the same for all $\boldsymbol{Y}_i$. The first equality in (7.9) is the definition of Fisher information and the second equality is a result that follows from one of the standard regularity conditions.

In a problem with independent but not identically distributed $\boldsymbol{Y}_i$ we might use $\mathcal{I}_n = n\mathcal{H}_n(\boldsymbol{\theta})$ as in item 2 previously. Here, $\mathcal{H}_n(\boldsymbol{\theta})$ would most likely be taken as the average Fisher information matrix $\mathcal{I}_F(\boldsymbol{\theta})$, the $q \times q$ matrix with jk^{th} element

$$i_{j,k} = \frac{1}{n^2}E\left(\sum_{i=1}^n \frac{\partial \ell_i(\boldsymbol{\theta}|\boldsymbol{Y}_i)}{\partial \theta_j}\sum_{s=1}^n \frac{\partial \ell_s(\boldsymbol{\theta}|\boldsymbol{Y}_s)}{\partial \theta_k}\right) = \frac{1}{n}E\left(-\sum_{i=1}^n \frac{\partial^2 \ell_i(\boldsymbol{\theta}|\boldsymbol{Y}_i)}{\partial \theta_j\,\partial \theta_k}\right) \tag{7.10}$$

Note that, if we used this same form with independent and identically distributed $\boldsymbol{Y}_i$ we would have

$$\begin{aligned} i_{j,k} &= \frac{1}{n^2} E\left(\sum_{i=1}^{n} \frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{Y}_i)}{\partial \theta_j} \sum_{s=1}^{n} \frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{Y}_s)}{\partial \theta_k}\right) \\ &= \frac{1}{n^2} \sum_{i=1}^{n} \sum_{s=1}^{n} E\left(\frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{Y}_i)}{\partial \theta_j} \frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{Y}_s)}{\partial \theta_k}\right) \\ &= \frac{1}{n^2} \sum_{s=1}^{n} d_{i,j} = \frac{d_{i,j}}{n}. \end{aligned}$$

If we chose to use $\mathcal{I}_n = \mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ as in item 3 previously, $\mathcal{J}_n(\boldsymbol{\theta}, \boldsymbol{Y})$ would be what is called the average observed information $\mathcal{I}_n^O(\boldsymbol{\theta}, \boldsymbol{Y})$, which is usually taken to be the $q \times q$ matrix with jk^{th} element

$$i_{j,k}^o = \frac{1}{n} \sum_{i=1}^{n} -\frac{\partial^2 \ell_i(\boldsymbol{\theta}|\boldsymbol{y}_i)}{\partial \theta_j \partial \theta_k}, \tag{7.11}$$

although an alternative form is sometimes used,

$$i_{j,k}^o = \frac{1}{n^2} \sum_{i=1}^{n} \frac{\partial \ell_i(\boldsymbol{\theta}|\boldsymbol{y}_i)}{\partial \theta_j} \sum_{s=1}^{n} \frac{\partial \ell_s(\boldsymbol{\theta}|\boldsymbol{y}_s)}{\partial \theta_k}. \tag{7.12}$$

In contrast to average Fisher information, which is the same as Fisher information for a single observation if the $\boldsymbol{Y}_i$ are identically distributed, average observed information will remain distinct even in this case. But it gives the same limiting distribution as the other two and provides a suitable alternative for the $\mathcal{I}_n$ of (7.1), particularly if the expected values in (7.9) prove difficult to evaluate. In the case of independent but not identically distributed $\boldsymbol{Y}_i$, choice of average Fisher information or average observed information is typically based on ease of computation; sometimes taking expectations can reduce the complexity of expressions and $\mathcal{I}_n$ will be taken as $\mathcal{I}_F(\boldsymbol{\theta})$ while in other cases the expectations prove intractable and $\mathcal{I}_n$ will be taken as $\mathcal{I}_n^O$.

7.4 Asymptotic Context Revisited

The example of the previous section assumed the asymptotic context of a repeating lattice (and also presumed maximum likelihood estimation). The context of a repeating lattice provides replications $\boldsymbol{Y}_i$; $i = 1, \ldots, n$ of the entire process for a fixed lattice size of size m. This greatly facilitates both the production of results giving asymptotic normality and, importantly, our ability to compute the necessary quantities involved in $\mathcal{I}_n$, whatever form has been chosen for this sequence of normalizing matrices. Specifically, if independent replicates of $\boldsymbol{Y}_i$ are available, then forms of $\mathcal{I}_n$ as average information (either expected or observed) are computed as exactly that, averages over the replicate data values $\boldsymbol{y}_i$, assumed to be realizations of the $\boldsymbol{Y}_i$.

The asymptotic context of an expanding lattice, however, provides no true replicates of the process, only a single process that is continually growing in size. Recall that in this setting, i indexes only a single location, not an entire lattice and n represents the size of that lattice. Here, not only does the development of theoretical results become more challenging, but the question of how one computes realized values of the quantities involved can complicate the application of those results. Both of these difficulties are typically approached by placing restrictions on the strength of dependence allowed in the process, a Result due to Comets and Janzura presented in the chapter on composite likelihood to follow providing a notable exception. Such restrictions on the strength of dependence may be represented in terms of what are called mixing conditions, which imply essentially that two finite subregions of the lattice approach independence as the distance between them increases (cf., Gaetan and Guyon 2010, Appendix B.2). A mixing condition is often assumed as part of the conditions attached to a theoretical

result. But there is also a practical side to this. If the approach to independence implied by a mixing condition is "fast enough," then sufficiently separated subregions of a lattice could be considered as (nearly) independent replicates of the process if each of those subregions are, at the same time, large enough to reflect the dynamics of the entire process. This, of course, renders the entire idea impractical unless one has either a huge lattice or the process exhibits almost no dependence at all. Fortunately, it is not necessary that regions large enough to capture the dynamics of the entire process be (nearly) independent. Mixing conditions similar to those that allow the development of asymptotic normality produce the same type of convergence in averages familiar from the law of large numbers (i.e., averages are consistent for their expected value). What is really needed in an application, then, is to be able to define subregions or blocks, that may be overlapping, in such a way that (1) the blocks are large enough to each reflect the spatial structure of the entire process, and (2) the blocks are small enough that many replicates are available over which to take averages. This is the essence of methodologies collectively known as spatial subsampling. The interested reader is referred to, among others, Carlstein (1986), Kunsch (1989), Sherman (1996), Heagerty and Lele (1998), Nordman and Lahiri (2003, 2004), and Politis, Romano and Wolf (1999). The implementation of spatial subsampling to produce estimated values for the covariance matrix of a limiting normal distribution is an area in need of additional investigation. Lee and Lahiri (2003) and Nordman and Caragea (2009) examine the issues involved in the context of spatial subsampling for the estimation of variograms in a geostatistical setting.